

Exact oracle

- 128,881 comparisons
- 7 topology-edge settings
- 0 discrepancies

Deliberate faults

- 10/10 fault classes detected
- 280,216 fault-challenge comparisons
- 118,916 differentiating comparisons

128,881/128,881
exact matches

Convergent evidence

Exactness
Fault sensitivity
Practicality
Empirical concordance

Published data

- 877/877 BRANCHSNV unambiguous calls matched SNPPar
- 46/46 published focal-branch SNVs reproduced across 3 analyses
- 66 additional SNPPar events retained as placement-ambiguous

Scalability

- 39/39 CLI runs completed
- 2,000 taxa × 10,000 sites
- 250 taxa × 100,000 sites